PDHU - Mismatch of Vision and Affordability

ADE's Heat Networks division published its "Connecting the Pipes" report this week, setting out a joined-up vision for delivering clean heat by 2040. The report is well constructed, the goals are clear, and the institutional analysis on the need for Cabinet Office coordination, HM Treasury rebalancing of electricity levies, and Ofgem reform sets the focus.

The Pimlico District Heating Undertaking operated by Westminster City Council (WCC) provides a live, documented, and data rich case study of that divergence. Working with residents on the technical and regulatory challenge to Westminster City Council's proposed £150–200m decarbonisation programme which would impose charges of £44,000–£66,000 per flat on approximately 3,300 households with 50% being leaseholders I have spent the past year mapping the gap between what the sector says and what the funding architecture actually delivers.

This article examines several structural problems that the ADE report does not resolve.

1. Seventy-four years of history unsuited for proposed funding models

PDHU was designed following a 1946 feasibility study recommending use of waste heat from Battersea Power Station that generated electricity as a by-product of supplying heat to Churchill Gardens. The heat source was effectively free, - surplus thermal energy from electricity generation that would otherwise have been discharged into the Thames. Battersea Power Station had been conceived in 1927 and became operational from 1933. The pipework and accumulator were designed for the specific station configuration, operating at moderate flow temperatures consistent with back-pressure turbine exhaust, not dedicated boiler firing.

The original engineering paper, presented to the Institution of Civil Engineers in 1952–53, made five conclusions based on 24 months of actual operating data. The ratio of electricity to heat output — R — was measured at 0.17 meaning that for every unit of heat delivered to consumers, 0.17 units of electricity were generated, effectively subsidising the heat cost. The paper's conclusion was explicit, - combined heat and electricity generation "would give the greatest economy," but *"Britain could not afford district heating if it were to be a question of burning fuel in a glorified central-heating boiler."*

Battersea A closed in 1975, but the PDHU back-pressure turbines in Battersea B continued supplying free waste heat until Battersea B was decommissioned in 1982. Westminster City Council had anticipated the closure by constructing a replacement coal-fired boiler house on the power station site in 1981-82. Coal firing continued until 1988, when the boiler house was converted to gas.

Non-exhaustive timeline of PDHU-related activities is presented below:

YEAR	EVENT	HEAT SOURCE
1946	Feasibility study recommends Battersea waste heat	Free Battersea Power Station waste heat
1950	First heat to Churchill Gardens	Free Battersea waste heat (Battersea B)
1951	Dolphin Square connected	Free Battersea waste heat
1952	Russell House + Abbots Manor A connected	Free Battersea waste heat
1972	Lillington and Longmoore Gardens completed	Own oil-fired boiler house (7.25MW)

YEAR	EVENT	HEAT SOURCE
1975	Battersea A closes	Battersea B
1976	Original Darwin House built	Connected to PDHU
1981–82	New coal boiler house built	Coal (30 MW)
1982	Battersea B decommissioned	Coal boilers substituted
1983	Lillington & Longmoore Gardens connected to PDHU	PDHU - Coal boilers only — never free heat
1988	Move to Churchill Gardens and conversion to gas	Gas boilers 10.1MW each
1998–2000	Dolphin Square disconnects from PDHU	Builds 3 independent gas boilerhouses
2006	Abbots Manor B permanent PDHU connection	PDHU
2006	New PDHU Energy Centre: 3×8MW gas + 2×1.55MWe CHP	Gas
2006	Tachbrook Triangle connected	PDHU
2017	Over 100 dwellings of Peabody X, Y, Z and Wolfson House get connected to PDHU	PDHU
2021	PDHU CHP decommissioned	
2025	Redeveloped Darwin House reopens	Block-level ASHP — not reconnected to PDHU
2026	Proposed renewal levy	£150–200m programme

By 2026 all five of the original paper's engineering conclusions became invalid:

1. Heat is no longer cost-competitive with the proposed levy being approximately 5 times the cost of other heat network schemes based on the DESNZ dataset. WCC itself chose the block-level ASHP at the point of redeveloping Darwin House located 50 metres from PDHU energy centre.
2. Of the 50 GWh generated, only approximately 44 GWh reaches buildings against the original design target of near-zero distribution losses. The losses in the secondary in-block infrastructure or useful heat consumption cannot be quantified.
3. R has been permanently zero since 1982, - the electricity generation that the accumulator was specifically designed to optimise alongside heat supply no longer exists.
4. The accumulator, described as "essential to the economy of a combined heat-electric system," has no combined heat-electric system left to serve.
5. The coal-saving and atmospheric pollution benefits that provided the scheme's original public health justification have been replaced by 12,000–16,000 tonnes of CO_2 per year from gas boilers commissioned by the Westminster City Council after the closure of Battersea Power Station, now proposed to be solved at residents' expense.

2. The heat network sector outlier

The ADE cost reduction annex models heat networks using air-source heat pumps as the primary heat source, a COP of 2.7, a capacity factor of 37%, and capital costs of £1.3m/GWh based on the Green Heat Network Fund average. These are reasonable parameters for a well-run new-build network. They are not a description of the existing installed base.

The UK currently has approximately 500,000 heat network consumers across around 14,000 networks — an average of ca.36 consumers per network. The existing sector is almost entirely communal block-level systems, - a single boiler serving a low-rise or mid-rise residential building, probably gas-fired, probably without individual metering, and without or with minimal thermal storage and flexibility infrastructure the ADE cost model assumes.

PDHU serves 3,300 consumers across four estates with over 70 residential blocks and 50 commercial premises. It is approximately 90 times the average UK network by consumer count. It generates approximately 50,000 MWh per year from approximately 60,000 MWh of gas or 0.5% of government's annual UK heat network output estimate of 10 TWh. Of the 50 GWh generated, approximately 44 GWh reaches buildings after accumulator and primary network losses, with further unquantified secondary losses within buildings. It does not use flat-level metering despite Lillington Gardens Phase 1 blocks constructed in the 1960s being equipped with them. While the Heat Network (Metering and Billing) Regulations 2014 mandate block-level metering, not all blocks are equipped with them. Irrespective of the block-level metering presence, only estate-wide data is being used for billing. PDHU proposed renewal programme costs roughly £5–7m/GWh against ADE's own GHNF benchmark of £1.3m/GWh, 3-4 times the benchmark before professional fees.

Professional fees on the PDHU programme are approximately 30% of total programme cost. That exceeds the entire five-year national HNES capital budget that ADE is calling for to fix legacy networks across England.

The point is not that PDHU is uniquely poorly managed, though there are serious questions on that front. The point is that a 76-year-old district heating system, whose original free heat source closed 44 years ago, serving multiple mixed-tenure estates in inner London, is categorically different from the sector the ADE report models. Every funding instrument currently available was designed for the latter, and not the former.

3. The GHNF cap and estimates

The Green Heat Network Fund has two binding constraints on grant size. The grant cannot cover over 50% of eligible construction costs. And the total award cannot exceed 4.5 pence/kWh of heat **delivered to end customers** over the first 15 years of operation.

The GHNF guidance is unambiguous, - the application form divides the grant requested by "the total thermal energy delivered over the first 15 years of operation, - heat delivered to end customers." Not fuel consumed at the energy centre. Not heat generated by the boilers or heat pumps. Heat physically arriving at the point of consumption after all distribution losses.

This distinction matters for PDHU because the heat delivered to connected buildings is determined by those buildings' demand — it does not change with network renewal. Whether the network is efficient or degraded, the buildings require the same useful heat. Renewal reduces how much fuel must be burned to meet that demand — a genuine benefit in terms of cost and carbon — but it does not increase the

delivered heat denominator in the GHNF calculation. The grant ceiling is set by what the estates need, not by how efficiently the network delivers it.

The measured energy flow

The full energy flow for the current PDHU network, derived from Loughborough University's peer-reviewed measured analysis of the same network (Martin-Du Pan et al., 2015) and PDHU's own calorifier instrumentation, is as follows:

- Fuel input at energy centre: ~60 GWh/year
- Heat generated: ~50 GWh/year (boiler seasonal efficiency ~83%, consistent with Loughborough's measured 84.3%)
- Accumulator losses: ~1 GWh/year (measured from PDHU instrumentation)
- Primary network losses (energy centre to building boundary): ~5 GWh/year (~10% of generated heat) — measured by Loughborough University
- **Heat delivered to buildings: ~44 GWh/year**
- Secondary losses within buildings: estimated at ~30% — unverifiable without flat-level metering which PDHU does not have considering the current metering regulation, those are not required given the legacy system and the fact the metering is uneconomic and installation of heat allocators is no longer permitted
- **Heat consumed by residents or useful heat that includes DHW and SH - approximately 31 GWh/year, - estimated, not measured.**

Of the approximately 44 GWh delivered to buildings, space heating accounts for approximately 21.3 GWh per year, as documented in WCC's OBC technical documentation. Domestic hot water accounts for the remainder after secondary losses. With secondary losses estimated at ~30%, total consumer demand is approximately 31 GWh/year, of which space heating is 21.3 GWh and DHW approximately 9.7 GWh — roughly 31% of total consumer demand, or approximately 46% of space heating demand. This DHW figure is consistent with typical per-dwelling consumption benchmarks for the connected population of approximately 3,300 dwellings. WCC's own OBC technical documentation explicitly assumes domestic hot water demand remains unchanged pre and post upgrade works — only space heating reduces with fabric improvements, from 21.3 to 18.3 GWh/year. Since DHW anchors approximately 31–40% of total consumer demand regardless of fabric improvements, the GHNF denominator is relatively stable across scenarios. All secondary loss figures and resulting consumer demand estimates are uncertain and unverifiable without collecting any additional energy consumption and heat loss data. None are available since the PDHU decarbonisation project evaluation started in 2021.

The primary loss figure challenged and not corrected

WCC's OBC technical documentation used a primary network loss assumption of approximately 9.3 GWh — equivalent to 18.6% of generated heat, nearly double the 10% measured by Loughborough University on the same network. This assumption was formally challenged by the Resident Technical Working Group with reference to the peer-reviewed research. The challenge was not responded to. The assumption remained unchanged in the January 2025 revision of WCC's technical documentation, at 75% confidence.

The practical consequences of overstating primary and other losses are direct, - the heat source is sized to cover losses that measured data does not support; the primary network replacement is specified against highly uncertain heat loss rate. The project fees, optimism bias assumptions are calculated as a percentage of total programme cost.

The GHNF cap applied to PDHU's measured energy flows

The 44 GWh currently delivered to buildings is the correct and measurable GHNF denominator and it is the only figure in the loss chain that can be verified without acquisition of the additional energy consumption and heat loss measurements and calculations. Post-renewal, primary losses reduce from ~5 GWh to ~1.1 GWh and secondary losses from ~30% to an estimated ~20% potentially reducing useful heat demand to under 40 GWh subject to feasible replacement of secondary in-block piping and space availability for installation of required installation.

Scenario	Fuel	Generated	Calorifier	Primary Loss	To Buildings	15yr	4.5p Max	2p Realistic
Current (measured)	~60 GWh (gas)	~50 GWh	~1 GWh	~5 GWh (10%)	~44 GWh	660,000 MWh	£29.7m	£13.2m
Post-renewal	~16-18 GWh (elec)	~40-41 GWh	~1 GWh	~1.1 GWh (2.9%)	~38-40 GWh	570-600,000 MWh	£25.7-27m	£11.4-12m
WCC OBC assumed (inflated primary)	~60 GWh (gas)	~50 GWh	~1 GWh	~9.3 GWh (18.6%)	~40 GWh	600,000 MWh	£27m	£12m
Small new-build block (~5% losses)	—	~1,050 MWh	—	~50 MWh	~1,000 MWh	15,000 MWh	~£675k	~£300k

Sources: WCC OBC technical documentation presented to residents, CEAH and H&S Scrutiny Committees and Cabinet meetings, 2025; Martin-Du Pan, O. et al. (2015), 'Current and future operation scenarios for a 50,000 MWh district heating system', Architectural Engineering and Design Management, 11(4), pp. 280–304; GHNF Guidance for Applicants R11, August 2025.

Post-renewal, primary losses reduce from ~5 GWh to ~1.1 GWh and secondary losses from an estimated ~30% to ~20%, reducing heat delivered to buildings to approximately 38-40 GWh, subject to feasible replacement of secondary in-block pipework and space availability for insulation installation. The GHNF denominator therefore falls slightly post-renewal rather than remaining constant, as lower secondary losses mean less heat needs to be supplied to meet the same resident demand of approximately 31 GWh. The grant ceiling moves correspondingly:

- Current network (44 GWh to buildings) at 15-year delivery 660,000 MWh → 4.5p/kWh maximum **£29.7m** → 2p/kWh competitive **£13.2m**
- Post-renewal (38-40 GWh to buildings) at 15-year delivery 570-600,000 MWh → 4.5p maximum **£25.7-27m** → 2p competitive **£11.4-12m**

The reduction in ceiling from current to post-renewal is modest at ca. £2-4m and does not materially change the fundamental position. In all scenarios the maximum realistic grant represents approximately 6-14% of a £150-200m programme cost.

WCC's OBC technical documentation assumed a realistic competitive grant of 2p/kWh and not the 4.5p/kWh absolute maximum without any explanation why the ceiling is not achievable. The documentation is explicit that this competitive rate "is by a considerable margin the limiting factor which caps grant support." Applied to WCC's own delivered heat estimate of approximately 35 GWh and using

the uncertain secondary-loss-adjusted consumer demand figure rather than the building boundary figure this gives approximately £10.5m, consistent with the £10m figure presented to residents. Applied to the building boundary delivery of 44 GWh being the only figure supported by measured data with the same 2p/kWh rate giving £13.2m. The absolute maximum at 4.5p/kWh is £29.7m on current measured figures, falling to approximately £25.7-27m post-renewal as secondary losses reduce.

Neither figure makes the programme viable without overwhelming leaseholder cost recovery. The key eligibility assumption that secondary pipework and HIUs qualify as eligible costs is held at only 40% confidence in WCC's own technical documentation. What is clear is that narrowing the eligible cost base to the heat source alone, - the approach presented by WCC to residents would reduce the grant ceiling substantially.

There is a further issue embedded in WCC's technical modelling, - the energy model includes a demand reduction from 21.3 to 18.3 GWh/year for space heating based on fabric improvement works discussed at a spring 2024 workshop, held at 50% confidence. Domestic hot water demand is explicitly assumed unchanged. No funded fabric improvement programme exists for PDHU-connected buildings while the £43m Climate Works budget for the next 5 years covers non-PDHU stock only. The demand reduction assumption underpinning the heat source sizing, network design, and grant calculations has no funded delivery mechanism or committed funding behind it.

For a small new-build single block network, the grant is limited by the 50% of capex rule and not the per-kWh ceiling, which would allow a larger amount. For PDHU, the two limits interact depending on how broadly GHNF defines eligible costs. If the full secondary network and HIUs qualify, the per-kWh ceiling of approximately £27-30m is the lower of the two limits and determines the grant. If only the heat source and primary network qualify, the 50% of eligible costs rule produces the lower figure at approximately £14-20m. In either case the maximum grant represents approximately 7-15% of a £150-200m programme. The fund was designed for the future sector. It does not reach the existing one at this scale. To maximise GHNF availability and reduce primary network related capex and associated heat losses, a block-by-block funding approach would merit serious consideration, provided there are no technical limitations to block-level solution implementation. It is not clear this has been systematically assessed.

4. The HNES gap as architecture gap

The ADE report calls for £5bn (also confirmed at the evidence session of the GLA Environment Committee on 4 Nov 2025) through the Heat Network Efficiency Scheme to ensure that legacy network improvements do not fall solely on vulnerable consumers. This position is also reinforced by the government's Heat Network Zoning consultation response of January 2026, which stated that leaseholders in existing communally heated buildings within designated heat network zones should face connection and upgrade costs capped at zero. The ADE framing is correct but concerns and doubts around its delivery remain.

HNES's total budget across its initial deployment was £32m. It cannot fund heat source replacement, only efficiency improvements to existing distribution infrastructure. PDHU's renewal programme at £150–200m would consume five to six times the entire national HNES budget. The secondary and tertiary network works alone, the portion most directly relevant to HNES's stated scope are estimated at £100–120m across the four estates.

HNES is correctly sized for what the average 36-consumer block network needs, - an optimisation study, some pipework insulation and controls upgrade. It is structurally unsuited to PDHU by an order of magnitude. What is needed for networks at PDHU's scale is something closer to a regulated asset base or

a dedicated legacy district heating fund with a separate grant ceiling structure and not a scaled-up version of a scheme designed for boiler room optimisation.

5. Fabric-first sequencing requirements ignored

The Warm Homes Plan, CIBSE CP1, and GHNF's own competitive scoring criteria all point toward the same sequencing: reduce demand through fabric improvements first, then right-size and decarbonise the heat supply system.

The logic as it applies to GHNF funding is precise. The 4.5p/kWh cap is expressed per unit of heat delivered to end customers. Fabric improvements reduce demand, which reduces both required network capacity and programme capex. Because capex falls proportionally faster than delivered volume, - smaller heat source, smaller pipe diameters, simpler HIU, the grant-per-kWh ratio improves even as the absolute ceiling falls. A fabric-first network also runs at lower flow temperatures, improving heat pump COP, reducing distribution losses, and lowering carbon intensity per delivered kWh. All three improve GHNF carbon gate performance and competitive scoring, - the carbon abatement adjustment metric alone is worth up to 30% of a project's competitive score.

The interaction between fabric improvements and distribution losses is particularly important. Lower flow temperatures made possible when buildings are properly insulated reduce the temperature differential between primary pipework and the surrounding ground or ambient temperature for the secondary pipework. Distribution losses are directly proportional to that differential. Fabric-first therefore reduces losses as well as demand, improving the delivered kWh denominator in the GHNF cap calculation even relative to a network serving lower absolute demand.

Westminster City Council's selective licensing scheme in place from November 2025 requires minimum energy efficiency standards across the borough including properties where WCC itself is the freeholder. The PDHU-connected estates fall within this requirement. Yet the £43m Climate Works programme delivering insulation and fabric improvements covers non-PDHU stock only. The 3,300 households captive to PDHU have received no equivalent fabric investment leaving WCC potentially non-compliant with its own licensing standards in the very properties it is proposing to charge £44,000-66,000 each for heat network renewal.

Well-insulated 1950s–1960s social housing of the type found across PDHU's estates could see 30–40% heat demand reduction through fabric improvements and double glazing. Correctly sizing the network to that reduced demand would mean a materially lower programme capex producing a GHNF application stronger on every competitive metric, at lower total cost, with a smaller residual gap to be funded from borrowing and leaseholder charges.

Instead, WCC is proposing to renew at full current-demand scale, locking in oversized infrastructure for 25–40 years at maximum cost, while simultaneously making the GHNF application weaker than it would be after fabric-first sequencing. The programme cost reduction from right-sizing could easily exceed the grant foregone by the lower delivered kWh denominator meaning fabric-first is not just better for residents' comfort and bills, it is arithmetically better for the funding ratio.

The policy hierarchy that should have governed this programme, - fabric first, then right-size, then decarbonise is stated in the Warm Homes Plan, embedded in CIBSE CP1, rewarded by GHNF's scoring criteria, and acknowledged by ADE's own theory of change. It was not followed. The people paying for that choice are the residents.

6. Undisclosed carbon reduction beneficiary

PDHU generates approximately 50,000 MWh of heat per year — roughly 0.5% of UK government's estimate for annual heat network sector output of 10 TWh. Yet its renewal programme would consume five to six times the entire national HNES budget. The funding instruments don't reach it not because it is unusual, but because it is large in a sector built around small.

Westminster City Council voted in September 2020 to become carbon neutral by 2030. PDHU accounts for approximately 39% of the Council's total carbon emissions, approximately 12,000–16,000 tonnes CO_2 per year. Decarbonising PDHU is therefore not primarily a consumer benefit, - it is a council target compliance mechanism. WCC's own OBC and policies explicitly acknowledge this, stating that "carbon reduction is not the primary objective of this programme, which is to ensure reliable and affordable heat and hot water." Yet the carbon reduction is the primary justification for the scale and urgency of the programme, and the primary reason GHNF funding is available at all. Those carbon emissions, it bears repeating, are emissions WCC created by converting to gas boilers after Battersea closed, - decision residents were never consulted despite being made post right-to-buy.

The carbon arithmetic is instructive. Full decarbonisation of PDHU might reduce emissions by approximately 10,000 tonnes CO_2 per year — roughly a 90% reduction. At the HMT Green Book appraisal value of £265 per tonne, - the correct metric for public sector project appraisal under Treasury guidance, and the methodology WCC is required to use in its own OBC, - that reduction is worth approximately £2.65m per year in social value terms. Over a 25-year asset life, the total carbon benefit of PDHU decarbonisation is approximately £66m (more for 40-year period used by WCC) in present value terms at Green Book values.

Not one penny of that carbon benefit has been articulated as flowing back to the consumers funding the programme. Westminster City Council will count the reduction against its 2030 neutrality target. It will appear in WCC's sustainability reporting. The leaseholders will receive a bill for £44,000–£66,000 each if paid within 5 years or over £100,000 with interest included when spread over 20-year period.

Under HMT Green Book methodology, the carbon benefits of a publicly funded programme should be attributed to the funders of that programme. If leaseholders along the WCC HRA budget are the primary funders of PDHU decarbonisation, then the carbon credit also belongs to them, not to the Council's corporate accounting position. At £265/tonne and 10,000 tonnes/year, the annual carbon value is £2.65m — approximately £800 per consumer per year. Over the 25-year asset life that represents an NPV of approximately £12,500 per consumer, which should appear as a credit against any renewal levy rather than as a windfall for the Council's carbon and corporate accounts.

Meanwhile Westminster is allocating £43m on Climate Works fabric improvements on its non-PDHU connected housing stock, - buildings that already have individual boilers and whose residents are not captive. The PDHU-connected estates, where fabric improvements would directly reduce heat demand, programme cost, and per-consumer levy, receive none of it.

The Council is proposing to spend hundreds of millions on a heat network renewal to meet its carbon targets while the buildings connected to that network remain insufficiently insulated, and then charging the captive consumers for the carbon benefits that accrue to the Council. Options have been ranked and shortlisted without disclosing to consumers any analysis of the impact on their heating and hot water bills under each scenario, or how proposed post-renewal tariffs compare with current charges. The carbon reduction beneficiary is clearly identified. The consumer cost beneficiary is not.

7. Three revealing decisions

Three recent Westminster City Council decisions, taken alongside the PDHU decarbonisation and renewal programme, reveal the framework governing its approach to heat decarbonisation more clearly than any policy document.

1. There are 42 blocks with gas-fired communal boilers across Westminster City Council emitting annually 14,700 tCO_2e and exceeding those 12,000tCO_2e of over 70 blocks connected to PDHU. In 2024, Westminster replaced at least one communal gas boiler in its housing stock of 24 dwellings with a like-for-like gas boiler. The building contains only supported accommodation (no leaseholders) meaning the full renewal cost falling on the Housing Revenue Account. Carbon compliance, urgent enough to pursue £44,000–£66,000 levy on PDHU leaseholders, was not urgent enough to justify a low-carbon alternative when the Council alone bore the cost that on average worked out at £10,000/dwelling.

2. Darwin House within Churchill Gardens estate, - immediately adjacent to the PDHU accumulator, the heat distribution hub at the centre of the network Westminster proposes to renew at £150–200m — was originally connected to PDHU when first built in 1976. When the building was demolished and redeveloped as new supported accommodation, Westminster City Council chose not to reconnect to PDHU. Block-level air source heat pumps were installed instead. Darwin House contains only supported accommodation with no leaseholders, meaning the full capital cost falls on the Housing Revenue Account.

3. Dolphin Square, - one of Europe's largest residential estates, and an original founding connection to PDHU from 1951 disconnected from the network between 1998 and 2000, constructing three independent gas boiler houses. It has operated independently for over 25 years. Following its subsequent acquisition, a comprehensive programme of window replacements, insulation and conversion to block-level air source heat pumps has been underway across the listed estate. Several towers already converted, with the remainder ongoing. Westminster's position that disconnection from PDHU is not a viable option for existing connected buildings is directly contradicted by the largest building of over 1,200 apartments previously connected, which has been self-sufficient for a quarter of a century and is now decarbonising through exactly the block-level ASHP.

The three decisions form a coherent pattern. When cost falls on the HRA alone, Westminster chooses the cheapest option regardless of carbon compliance. When cost falls on HRA for a new build adjacent to PDHU, Westminster chooses block-level ASHP. When a building with negotiating power leaves the network, Westminster permits it. When 3,300 households are captive and made up of 50% leaseholds that can be charged, Westminster proposes full network renewal at maximum scale and no upfront energy efficiency measures.

Could the presence of leaseholders and not carbon compliance, network condition, or technical feasibility be a guiding principle of decision making?!

8. What the sector needs to say

This analysis is offered in support of the sector's goals, not in opposition to them. ADE is right that the heat transition requires HM Treasury commitment, regulatory reform, strategic zoning, and institutional coordination. The vision in "Connecting the Pipes" is the right one.

But trust which the report correctly identifies as a prerequisite for the heat transition cannot be built while 3,300 households in Pimlico are being told that £44,000–£66,000 per flat is the price of a transition they didn't choose, in a network they've been captive to since 1950, with a funding architecture whose calibration means they receive the smallest share of available grant per pound of programme cost.

PDHU was designed around free energy. That energy source closed 44 years ago. The engineers who designed the system said in 1950 that district heating without combined heat and power generation was too expensive to justify. They were right. What is now proposed at leaseholders' expense is the decarbonisation and renewal of exactly that uneconomic configuration, at a cost structure 3-4x the sector benchmark, sized for unreduced demand in uninsulated buildings, with a grant programme whose delivered-kWh cap means the more heat the network loses, the less public funding it receives.

PDHU is not an outlier. It is what happens when there is no binding economic rationality threshold, no systematic fabric-first demand assessment before sizing a renewal programme, no requirement for a network optimisation study before committing to full renewal, no enforceable cost allocation framework, and a grant programme whose per-kWh cap measured in delivered heat, means the largest legacy networks at this scale receive the smallest share of available public funding as a proportion of programme cost.

Ofgem's heat network regulation went live in January 2026. The fair pricing framework is still under development. The Section 20 notices have not yet landed; the alternatives were eliminated despite HM Treasury Green Book requirements. There is still time for the policy architecture to catch up with the consumer reality but only if the sector is willing to name the gap rather than paper over it before examples like PDHU contribute to a complete loss of trust in local administration and derail initiatives in scaling up heat network share to 20%.

KEY REFERENCES USED IN THE DOCUMENT

- ADE "Connecting the Pipes: Clean Heat by 2040" (March 2026) — primary subject
- ADE Technical Annex on Cost Reductions (March 2026)
- Donkin, Margolis & Carrothers, "Pimlico District Heating Undertaking," ICE Proceedings, 1952–53 — original engineering paper
- WCC 'Pimlico District Heating Undertaking: A Brief History 1946–2013' — WCC published history, key dates throughout
- WCC Strategic Outline Case, PDHU Decarbonisation V1 (2023)
- WCC Outline Business Case Approach, assumptions and documentation (2024–25), CAEH Scrutiny Committee meetings
- WCC OBC technical documentation presented to residents, January 2025 (Assumptions Log)
- Martin-Du Pan, O. et al. (2015), 'Current and future operation scenarios for a 50,000 MWh district heating system', Architectural Engineering and Design Management, 11(4), pp. 280–304
- GHNF Guidance for Applicants R11 (August 2025)
- HMT Green Book 2022 — carbon appraisal values
- WCC HRA 30-Year Business Plan and Housing Investment Plan 2024–25
- WCC HRA Capital Programme Overview
- Heat Network (Metering and Billing) Regulations 2014 (amended 2020)
- Ofgem Heat Network Authorisation Conditions AC A1 and AC A9 (2026)